

1. A pilot of a normally-aspirated, twin-engine airplane departs out of Rocky Mountain Metropolitan airport (KBJC) in the morning on a nice summer day and flies into Aspen/Pitkin County airport (KASE).
2. The pilot enjoys the scenery around Aspen, eats lunch, and decides to return home in the early hot afternoon.
3. The pilot departs KASE off of runway 33. At full throttle/power the airplane takes longer to accelerate but rotates at the normal speed.
4. The pilot pitches to the normal pitch target, retracts the gear, and initiates a climb.
5. The pilot notices the airplane isn't performing as desired. The pilot checks to see if the gear is up and adjusts the mixtures to try and get a little more power.
6. The terrain is rising, the pilot gradually pitches up, and the airplane starts losing airspeed.
7. The airplane quits climbing.
8. The stall horn begins to sound.

The above scenario is hypothetical, but there have been very similar situations that have ended tragically.

The airplane in the above scenario has encountered an unintended deceleration and impending sink rate that could rapidly become irreversible. This can be shown in two ways, using the traditional power curve [Figure 4-14] and the energy map [Figure 4-15]:

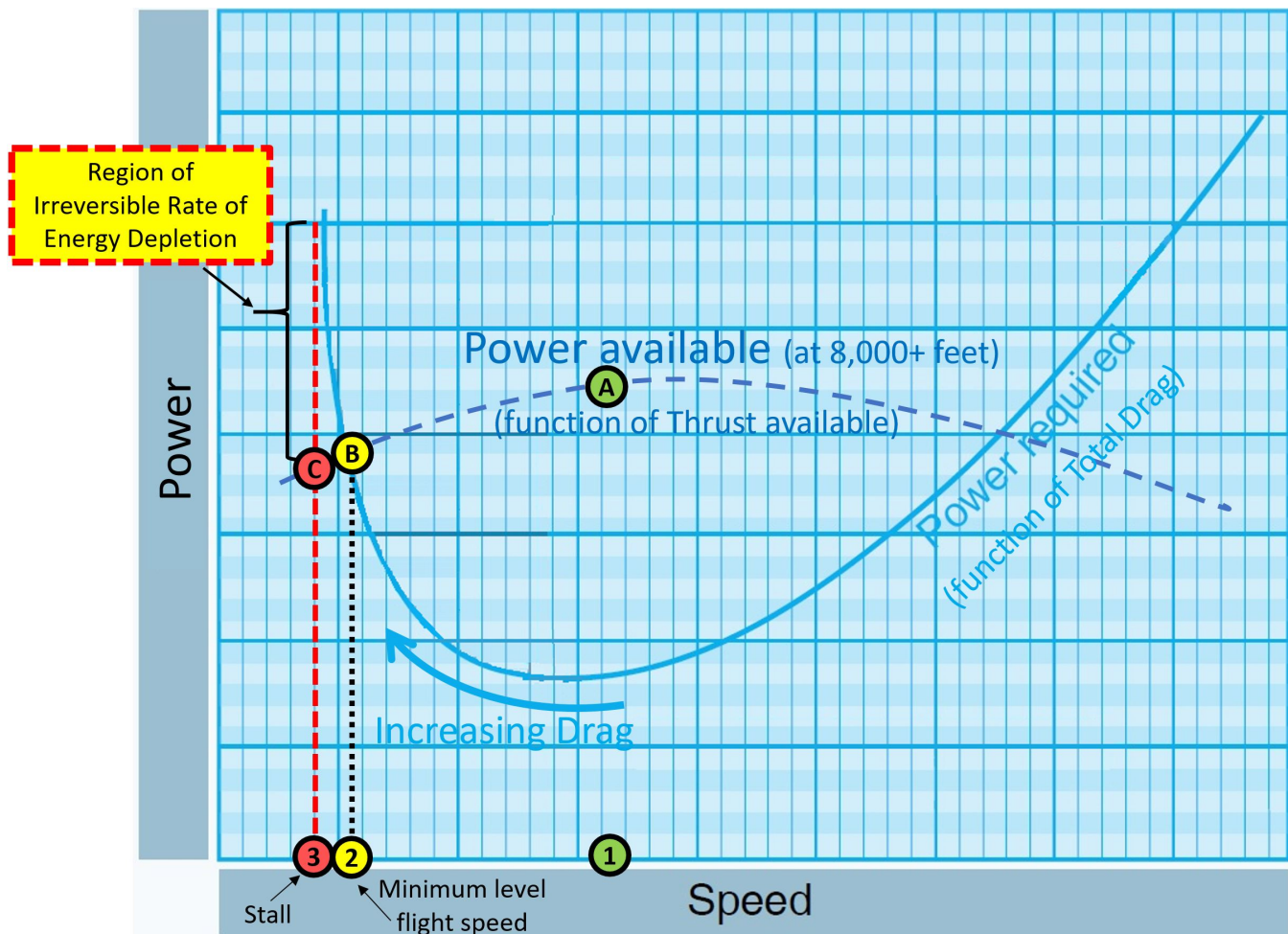


Figure 4-14. The energy depletion scenario viewed in the power required and available curve. Compared with the power available curve depicted in Figure 4-4, note the lower power available curve at this high elevation (7,820 feet at the departure airport) and higher density altitude than standard during a hot afternoon.

As illustrated in the airplane's power required and available curves [Figure 4-14], the airplane slows down, going from speed 1 where it is climbing (A: power available greater than power required), to speed 2, where it stops climbing (B: power available equal to power required), and continuing to speed 3 where the stall horn sounds (C: power available less than power required). The energy map [Figure 4-15] tells the same story from a total mechanical energy standpoint: the airplane has positive P_s at point 1 and climbs to point 2 where it stops climbing since $P_s = 0$, then continues to point 3, where the $P_s < 0$ and the stall horn sounds.

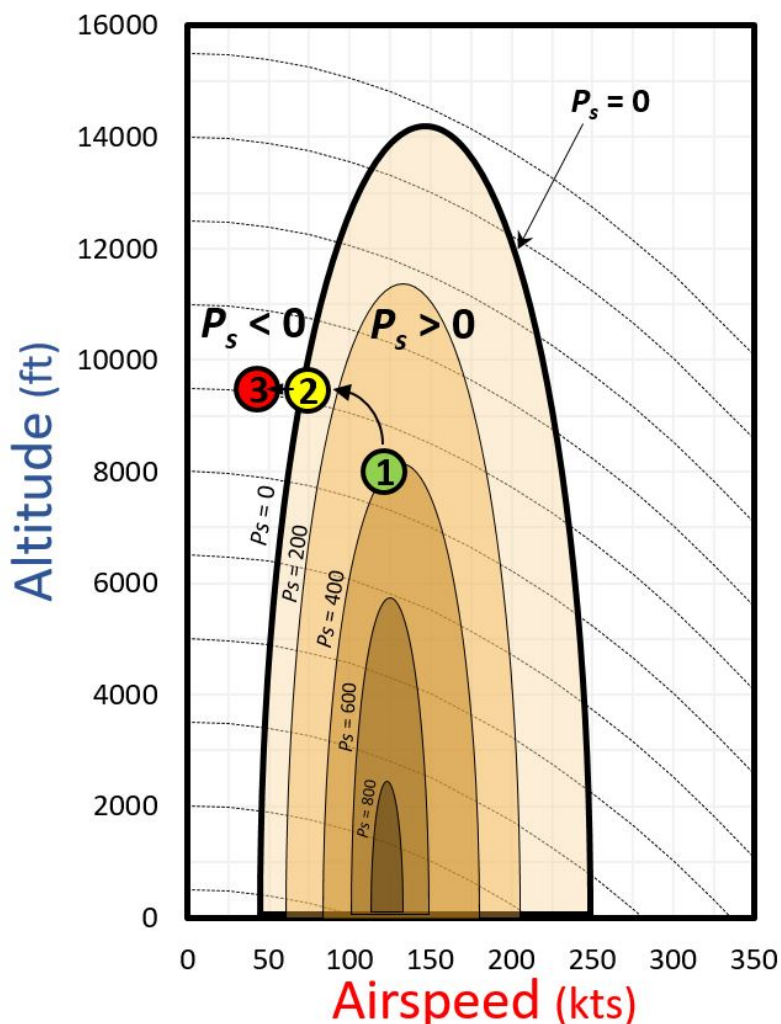


Figure 4-15. The energy depletion scenario viewed in the energy map. Specific excess power (P_s) contours are labeled in units of feet per minute.

The question then is: what does the pilot do to recover from this predicament? The answer is proper energy management. The airplane needs to move to a different place on the energy map that will allow the airplane to begin climbing. So, what does that mean?

- As can be seen in Figure 4-12, the pilot is in a scenario akin to that at the desired altitude, but with cautions when very slow.
- The pilot then has to do something that is not intuitive; consider gaining speed at the expense of some altitude initially to improve climbing performance with full throttle.
- Once the airplane accelerates to an airspeed in which the $P_s > 0$, it can begin to climb again.

The above recovery scenario is shown in the energy map Figure 4-16, which illustrates the important role of the elevator in assisting the pilot to recover from unintentional and dangerous deceleration and/or sink rate (refer to *Additional Role for the Elevator* section).

The airplane needs to gain speed at the expense of some altitude, moving from point 3 where the $P_S < 0$ to point 4 where the $P_S > 0$. The airplane can then initiate a constant airspeed climb to point 5, at the desired target altitude and airspeed [Figure 4-16]. Note that the desired target climb airspeed in the presence of rising terrain may be V_X , the speed for best angle of climb. V_X is slightly slower than V_Y , the speed for best rate of climb, and will result in a lower climb rate but steepest climb angle. Once the airplane has recovered from the unintentional airspeed loss and begins climbing at V_X , the pilot should assess the situation and make an important decision to mitigate further risk—either continue climbing or do something else. Should the airplane not have the needed performance to safely clear the rising terrain on its intended course, the pilot has at least another available option: make a 180 degree turn and return to land at the departure airport until temperature and density altitude conditions improve.

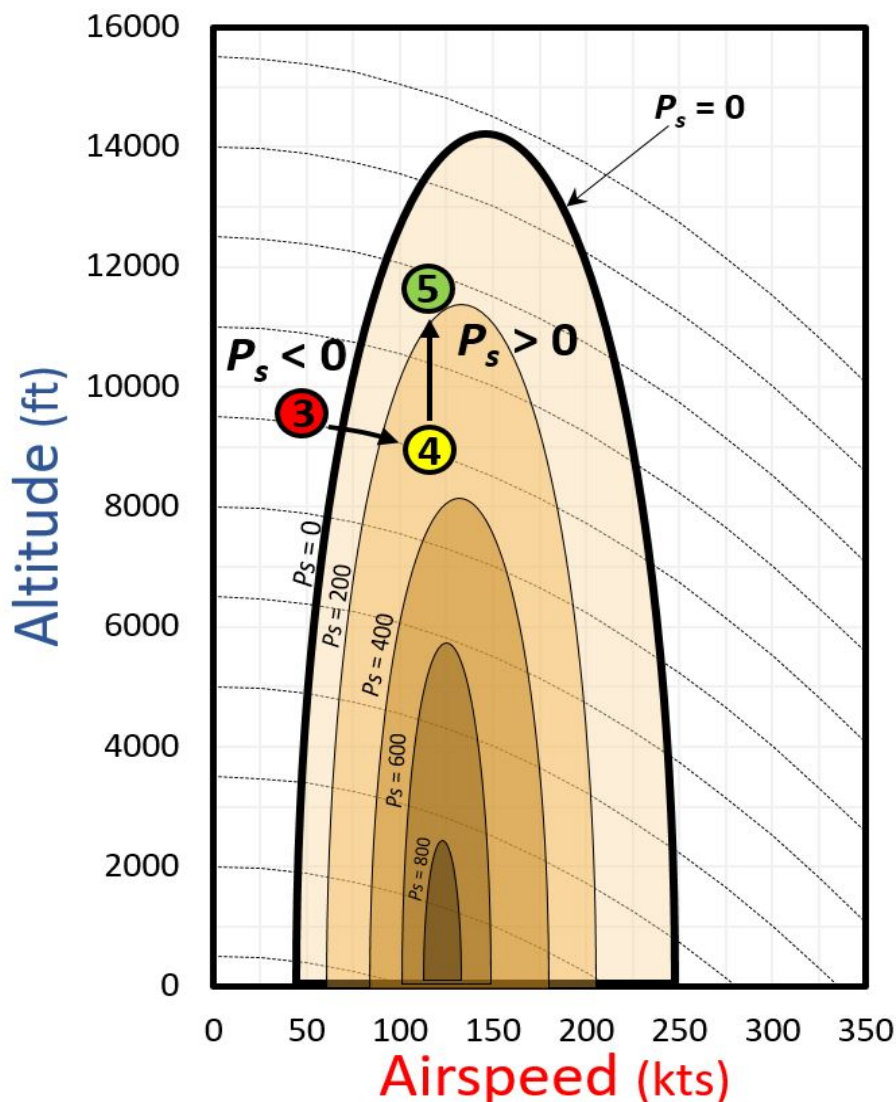


Figure 4-16. The energy loss scenario recovery viewed in the energy map. Specific excess power (P_S) contours are labeled in units of feet per minute.

The above rising terrain scenario is just one example illustrating the risk of irreversible deceleration and/or sink rate. Pilots need to be aware that unintentional depletion of mechanical energy can happen in various instances, especially as the airplane approaches the slow edge of its energy envelope at low altitude, where available specific excess power (P_S) is zero. Examples include unstable/slow approaches to landing; high-drag go-arounds where the pilot neglects to raise the gear and/or flaps; and steeper-than-normal turns in the traffic pattern. Note that irreversible sink rates do not necessarily involve exceeding the critical AOA resulting in a stall and spin. The airplane can be unstalled and still experience unrecoverable sink rates near the high-speed edge of its energy envelope, where available specific excess power (P_S) is also zero. Two examples are high-speed steep spirals following botched steep level turns, and high-speed dives too close to the ground.

The bottom line? Should the airplane ever experience unintended excessive negative energy rates with little or no excess power available under a given flight condition, the pilot needs to use proper energy management allowing a prompt recovery and a suitable follow-up action.

Review of Terms and Definitions

The terms and definitions specific to this chapter appear below.

Aircraft Energy Management

The process of planning, monitoring and controlling altitude and airspeed targets in relation to the airplane's energy state. Note that this definition is concerned with managing mechanical energy (altitude and airspeed) and addresses the safety (flight control) side of energy management. It does not address the efficiency (aircraft performance) side of energy management, which is concerned with how efficiently the engine generates mechanical energy from fuel and how efficiently the airframe spends that energy in flight.

Energy System

A flying airplane is an *open* energy system. That means that the airplane can gain energy from some source (e.g., fuel) and lose energy to the environment (e.g., surrounding air). In addition, energy can be added to or removed from the airplane's total mechanical energy stored as altitude and airspeed.

Total Mechanical Energy

Sum of the energy in altitude (potential energy) and the energy in airspeed (kinetic energy).

Kinetic Energy

Amount of energy due to the airspeed, expressed as $\frac{1}{2}mV^2$, where m = airplane's mass, and V = airspeed.

Potential Energy

Amount of energy due to the altitude, expressed as mgh , where m = airplane's mass, g = gravitational constant, and h = altitude.

Energy State

The airplane's total mechanical energy and its distribution between altitude and airspeed.

Energy Exchange

Trading one form of energy (e.g., altitude) for another form (e.g., airspeed).

Energy Balance Equation

According to this equation, the net transfer of mechanical energy into and out of the airplane (a function of thrust minus drag) is always equal to the change in its total mechanical energy (a function of altitude and airspeed). Note that this simplified definition does not account for long-term changes in total mechanical energy caused by the reduction in aircraft weight as fuel is gradually burned in flight.

Power Available

The airplane's rate of energy gain due to maximum available engine thrust at a given airspeed. Expressed as TV , where T = engine thrust and V = airspeed. Usually measured in horsepower, foot-pound per minute, or foot-pound per second.

Power Required

The airplane's rate of energy loss due to total drag at a given airspeed. Expressed as DV , where D = total drag and V = airspeed. Usually measured in horsepower, foot-pound per minute, or foot-pound per second.

Specific Excess Power (P_S)

Measured in feet per minute or feet per second, it represents rate of energy change—the ability of an airplane to climb or accelerate from a given flight condition. Available specific excess power is found by dividing the difference between power available and power required by the airplane's weight.

Energy Height or Total Specific Energy (E_s)

Measured in units of height (e.g., feet), it represents the airplane's total energy per unit weight. It is found by dividing the sum of potential energy and kinetic energy by the airplane's weight. It also represents the maximum height that an airplane would reach from its current altitude, if it were to trade all its speed for altitude.

Energy Error

An altitude and/or airspeed deviation from an intended target expressed in terms of energy. Depending on the airplane's total amount of energy and its distribution between altitude and airspeed, energy errors are classified as total energy errors, energy distribution errors, or a combination of both errors.

Total Energy Error

An energy error where the total amount of mechanical energy is not correct. The airplane has too much or too little total energy relative to the intended altitude-speed profile. When this error occurs, the pilot will observe that altitude and airspeed deviate in the *same* direction (e.g., higher and faster than desired; or lower and slower than desired). An example would be an airplane on final approach that is above the desired glide slope and at a faster airspeed than desired.

Energy Distribution Error

An energy error where the total mechanical energy is correct, but the distribution between potential (altitude) and kinetic energy (airspeed) is not correct relative to the intended altitude-speed profile. When this error occurs, the pilot will observe that altitude and airspeed deviate in *opposite* directions (e.g., higher and slower than desired; or lower and faster than desired). An example would be an airplane on final approach that is above the desired glide slope and at a slower airspeed than desired.

Irrecoverable Deceleration and/or Sink Rate

Unrecoverable depletion of mechanical energy as a result of continuous loss of airspeed and/or altitude coupled with insufficient excess power available under a given flight condition. Failure to recover above a certain critical AGL altitude results in the airplane hitting the ground regardless of what the pilot does.

Chapter Summary

Every pilot is an energy manager—managing energy in the form of altitude and airspeed from takeoff to landing. Proper energy management is essential for performing any maneuver as well as for attaining and maintaining desired vertical flightpath and airspeed profiles in everyday flying. It is also critical to flight safety since mistakes in managing energy state can contribute to loss of control in flight (LOC-I), controlled flight into terrain (CFIT), and approach and landing accidents. The objectives of this chapter are for pilots to: 1) gain an understanding of basic energy management concepts; 2) learn the energy role of the controls for managing the airplane's energy state; and 3) develop the ability to identify, assess, and mitigate risks associated with failure to manage the airplane's energy state.

Airplane Flying Handbook (FAA-H-8083-3C)

Chapter 5: Maintaining Aircraft Control: Upset Prevention and Recovery Training

Introduction

Safe pilots prevent loss of control in flight (LOC-I), which is the leading cause of fatal general aviation accidents in the U.S. and commercial aviation worldwide. LOC-I includes any significant deviation of an aircraft from the intended flightpath and it often results from an airplane upset. Maneuvering represents the most common phase of flight for general aviation LOC-I accidents; however, LOC-I accidents occur in all phases of flight.

To prevent LOC-I accidents, it is important for pilots to recognize and maintain a heightened awareness of situations that increase the risk of loss of control. Those situations include: uncoordinated flight, equipment malfunctions, pilot complacency, distraction, turbulence, and poor risk management. Attempting to fly in instrument meteorological conditions (IMC) when the pilot is not qualified or proficient is a common example of poor risk management. The *Emergency Procedures* chapter of this handbook contains specific information regarding unintended flight into IMC. Sadly, there are also LOC-I accidents resulting from intentional disregard for safety.

To maintain aircraft control when faced with these or other contributing factors, the pilot needs to be aware of situations where LOC-I can occur; recognize when an airplane is approaching a stall, has stalled, or is in an upset condition; and understand and execute the correct procedures to recover the aircraft.

Defining an Airplane Upset

The term “upset” was formally introduced by an industry work group in 2004 in the “Pilot Guide to Airplane Upset Recovery,” which is a part of the “Airplane Upset Recovery Training Aid.” The work group was primarily focused on large transport airplanes and sought to come up with one term to describe an “unusual attitude” or “loss of control,” for example, and to generally describe specific parameters as part of its definition. Consistent with the Guide, the FAA considers an upset to be an event that unintentionally exceeds the parameters normally experienced in flight or training. These parameters are:

1. Pitch attitude greater than 25°, nose up
2. Pitch attitude greater than 10°, nose down
3. Bank angle greater than 45°
4. Within the above parameters, but flying at airspeeds inappropriate for the conditions

The reference to inappropriate airspeeds describes a number of undesired aircraft states, including stalls. However, stalls are directly related to angle of attack (AOA), not airspeed.

To develop the crucial skills to prevent LOC-I, a pilot may receive academic or on-aircraft upset prevention and recovery training (UPRT), which should include: slow flight, stalls, spins, and unusual attitudes.

Upset training places considerable emphasis on understanding and preventing an upset, so a pilot avoids such a situation. If an upset does occur, upset training also reinforces proper recovery techniques. A detailed discussion of UPRT follows, including core concepts, what the training should include, and what airplanes or kinds of simulation can be used for the training. A discussion of various maneuvers and how to execute them follows later in this chapter.

Upset Prevention and Recovery

An unusual attitude is commonly referenced as an unintended or unexpected attitude in instrument flight. These unusual attitudes are introduced to a pilot during student pilot training as part of basic attitude instrument flying and continue to be trained and tested as part of certification for an instrument rating, aircraft type rating, and an airline transport pilot certificate. A pilot is taught the conditions or situations that could cause an unusual attitude, with focus on how to recognize one, and how to recover from one.

Unusual Attitudes Versus Upsets

Given the upset definition, there are a few key distinctions between an unusual attitude and an upset. An upset:

- Includes stall events.
- Includes overspeeds or other inappropriate speeds for a given flight condition.
- Has defined parameters. For example, for training purposes an instructor could place the aircraft in a 30° bank with a nose-up pitch attitude of 15° and ask the student to recover and that would be considered an unusual attitude, but would not meet the upset parameters.
- Centers on unintentional situations that may lead to a startle effect. For example, during unusual attitude training, the pilot is often directed to close their eyes, and any element of surprise disappears.

The top four causal and contributing factors that have led to an upset and resulted in LOC-I accidents are:

1. Environmental factors
2. Mechanical factors
3. Human factors
4. Stall-related factors

Environmental Factors

Turbulence, or a large variation in wind velocity over a short distance, can cause upset and LOC-I. Maintain awareness of conditions that can lead to various types of turbulence, such as clear air turbulence, mountain waves, wind shear, and thunderstorms or microbursts. In addition to environmentally-induced turbulence, wake turbulence from other aircraft can lead to upset and LOC-I.

Icing can destroy the smooth flow of air over the airfoil and increase drag while decreasing the ability of the airfoil to create lift. Therefore, it can significantly degrade airplane performance, resulting in a stall if not handled correctly.

Mechanical Factors

Modern airplanes and equipment are very reliable, but anomalies do occur. Some of these mechanical failures can directly cause a departure from normal flight, such as asymmetrical flaps, malfunctioning or binding flight controls, and runaway trim.

Upsets can also occur if there is a malfunction or misuse of the autoflight system. Advanced automation may tend to mask the cause of the anomaly. Disengaging the autopilot and the autothrottles allows the pilot to directly control the airplane and possibly eliminate the cause of the problem. For these reasons the pilot should maintain proficiency to manually fly the airplane in all flight conditions without the use of the autopilot/autothrottles.

Although these and other in-flight anomalies may not be preventable, knowledge of systems and AFM/POH recommended procedures helps the pilot minimize their impact and prevent an upset. In the case of instrument failures, avoiding an upset and subsequent LOC-I may depend on the pilot's proficiency in the use of secondary instrumentation and partial panel operations.

Human Factors

VMC to IMC

Unfortunately, accident reports indicate that continued VFR flight from visual meteorological conditions (VMC) into marginal VMC and IMC is a factor contributing to LOC-I. A loss of the natural horizon substantially increases the chances of encountering vertigo or spatial disorientation, which can lead to upset.

IMC

When operating in IMC, maintain awareness of conditions.

Diversion of Attention

In addition to its direct impact, an in-flight anomaly or malfunction can also lead to an upset if it diverts the pilot's attention from basic airplane control responsibilities. Failing to monitor the automated systems, over-reliance on those systems, or incomplete knowledge and experience with those systems can lead to an upset. Diversion of attention can also occur simply from the pilot's efforts to set avionics or navigation equipment while flying the airplane.

Task Saturation

The margin of safety is the difference between task requirements and pilot capabilities. An upset and eventual LOC-I can occur whenever requirements exceed capabilities. For example, an airplane upset event that requires rolling an airplane from a near-inverted to an upright orientation may demand piloting skills beyond those learned during primary training. In another example, a fatigued pilot who inadvertently encounters IMC at night coupled with a vacuum pump failure, or a pilot fails to engage pitot heat while flying in IMC, could become disoriented and lose control of the airplane due to the demands of extended—and unpracticed—partial panel flight. Additionally, unnecessary low-altitude flying and impromptu demonstrations for friends or others on the ground could lead pilots to exceed their capabilities, with fatal results.

Sensory Overload/Deprivation

A pilot's ability to adequately correlate warnings, annunciations, instrument indications, and other cues from the airplane during an upset can be limited. Pilots faced with upset situations can be rapidly confronted with multiple or simultaneous visual, auditory, and tactile warnings. Conversely, sometimes expected warnings are not provided when they should be; this situation can distract a pilot as much as multiple warnings can.

The ability to separate time-critical information from distractions takes practice, experience, and knowledge of the airplane and its systems. Cross-checks are necessary not only to corroborate other information that has been presented, but also to determine if information might be missing or invalid. For example, a stall warning system may fail and therefore not warn a pilot of close proximity to a stall, so other cues need to be used to avert a stall and possible LOC-I. These cues include aerodynamic buffet, loss of roll authority, or inability to arrest a descent.

Spatial Disorientation

Spatial disorientation has been a significant factor in many airplane upset accidents. Accident data from 2008 to 2013 shows nearly 200 accidents associated with spatial disorientation with more than 70% of those being fatal. All pilots are susceptible to false sensory illusions while flying at night or in certain weather conditions. These illusions can lead to a conflict between actual attitude indications and what the pilot senses is the correct attitude. Disoriented pilots may not always be aware of their orientation error. Many airplane upsets occur while the pilot is engaged in some task that takes attention away from the flight instruments or outside references. Others perceive a conflict between bodily senses and the flight instruments, and allow the airplane to divert from the desired flightpath because they cannot resolve the conflict.

A pilot may experience spatial disorientation or perceive the situation in one of three ways:

1. Recognized spatial disorientation: the pilot recognizes the developing upset or the upset condition and is able to safely correct the situation.
2. Unrecognized spatial disorientation: the pilot is unaware that an upset event is developing, or has occurred, and fails to make essential decisions or take any corrective action to prevent LOC-I.
3. Incapacitating spatial disorientation: the pilot is unable to affect a recovery due to some combination of:
(a) not understanding the events as they are unfolding, (b) lacking the skills required to alleviate or correct the situation, or (c) exceeding psychological or physiological ability to cope with what is happening.

For detailed information regarding causal factors of spatial disorientation, refer to Aerospace Medicine Spatial Disorientation and Aerospace Medicine Reference Collection, which provides spatial disorientation videos. The videos are available online at: www.faa.gov/about/office_org/headquarters_offices/avs/offices/aam/cami/library/online_libraries/aerospace_medicine/sd/videos/.

Surprise and Startle Response

Surprise is an unexpected event that violates a pilot's expectations and can affect the mental processes used to respond to the event. Startle is an uncontrollable, automatic muscle reflex, raised heart rate, blood pressure, etc., elicited by exposure to a sudden, intense event that violates a pilot's expectations.

This human response to unexpected events has traditionally been underestimated or even ignored during flight training. The reality is that untrained pilots often experience a state of surprise or a startle response to an airplane upset event. Startle may or may not lead to surprise. Pilots can protect themselves against a debilitating surprise reaction or startle response through scenario-based training, and in such training, instructors can incorporate realistic distractions to help provoke startle or surprise. To be effective the controlled training scenarios should have a perception of risk or threat of consequences sufficient to elevate the pilot's stress levels. Such scenarios can help prepare a pilot to mitigate psychological/physiological reactions to an actual upset.

Upset Prevention and Recovery Training (UPRT)

Upsets are not intentional flight maneuvers, except in maneuver-based training; therefore, they are often unexpected. The reaction of an inexperienced or inadequately trained pilot to an unexpected abnormal flight attitude is usually instinctive rather than intelligent and deliberate. Such a pilot often reacts with abrupt muscular effort, which is without purpose and even hazardous in turbulent conditions, at excessive speeds, or at low altitudes.

Without proper upset recovery training on interpretation and airplane control, the pilot can quickly aggravate an abnormal flight attitude into a potentially fatal LOC-I accident. Consequently, UPRT is intended to focus education and training on the prevention of upsets, and on recovering from these events if they occur. [Figure 5-1]



Figure 5-1. *Maneuvers that better prepare a pilot for understanding unusual attitudes and situations are representative of upset training.*

- Upset prevention refers to pilot actions to avoid a divergence from the desired airplane state. Awareness and prevention training serve to avoid incidents. Early recognition of an upset scenario coupled with appropriate preventive action often can mitigate a situation that could otherwise escalate into an LOC-I accident.
- Recovery refers to pilot actions that return an airplane that is diverging in altitude, airspeed, or attitude to a desired state from a developing or fully-developed upset. Recovery training serves to reduce accidents as a result of an unavoidable or inadvertently-encountered upset event. The pilot can learn to initiate a recovery to a normal flight mode immediately upon recognition of the developing upset condition. The pilot should ensure that control inputs and power adjustments applied to counter an upset are in direct proportion to the amount and rates of change of roll, yaw, and pitch, or airspeed so as to avoid overstressing the airplane unless ground contact is imminent.

UPRT Training Core Concepts

Airplane upsets are by nature time-critical events; they can also place pilots in unusual and unfamiliar attitudes that sometimes require counterintuitive control movements. Upsets have the potential to put a pilot into a life-threatening situation compounded by panic, diminished mental capacity, and potentially incapacitating spatial disorientation. Real-world upset situations often provide very little time to react, but exposure to such events during training can reduce surprise and mitigate confusion during an actual unexpected upset. The goal is to equip the pilot to promptly recognize an escalating threat pattern or sensory overload and quickly identify and correct an impending upset.

UPRT stresses that the first step is recognizing any time the airplane begins to diverge from the intended flightpath or airspeed. Pilots need to identify and determine what, if any, action should be taken. As a general rule, any time visual cues or instrument indications differ from basic flight maneuver expectations, the pilot should assume an upset and cross-check to confirm the attitude, instrument error or instrument malfunction.

To achieve maximum effect, it is crucial for UPRT concepts to be conveyed accurately and in a non-threatening manner. Reinforcing concepts through positive experiences significantly improves a pilot's depth of understanding, retention of skills, and desire for continued training. Also, training in a carefully structured environment allows for exposure to these events and can help the pilot react more quickly, decisively, and calmly when the unexpected occurs during flight. However, like many other skills, the skills needed for upset prevention and recovery are perishable and thus require continuous reinforcement through training.

UPRT in the airplane and flight simulation training device (FSTD) should be conducted in both visual and simulated instrument conditions to allow pilots to practice recognition and recovery under both situations. UPRT should allow them to experience and recognize some of the physiological factors related to each, such as the confusion and disorientation that can result from visual cues in an upset event. Training that includes recovery from bank angles exceeding 90 degrees could further add to a pilot's overall knowledge and skills for upset recognition and recovery. For such training, additional measures should be taken to ensure the suitability of the airplane or FSTD and that instructors are appropriately qualified.

Upset prevention and recovery training is different from aerobatic training. [Figure 5-2] In aerobatic training, the pilot knows and expects the maneuver, so effects of startle or surprise are missing. The main goal of aerobatic training is to teach pilots how to intentionally and precisely maneuver an aerobatic-capable airplane in three dimensions. The primary goal of UPRT is to help pilots overcome sudden onsets of stress to avoid, prevent, and recover from unplanned excursions that could lead to LOC-I.

Aerobatics vs. UPRT Flight Training Methods		
ASPECT OF TRAINING	AEROBATICS	UPSET PREVENTION AND RECOVERY TRAINING
Primary Objective	Precision maneuvering capability	Safe, effective recovery from aircraft upsets
Secondary Outcome	Improved manual aircraft handling skills	Improved manual aircraft handling skills
Aerobatic Maneuvering	Primary mode of training	Supporting mode of training
Academics	Supporting role	Fundamental component
Training Resources Utilized	Aircraft (few exceptions)	Aircraft or a full-flight simulator

Figure 5-2. *Some differences between aerobatic training and upset prevention and recovery training.*

Comprehensive UPRT builds on three mutually supportive components: academics, airplane-based training and, typically at the transport category type-rating training level, use of FSTDs. Each has unique benefits and limitations but, when implemented cohesively and comprehensively throughout a pilot's career, the components can offer maximum preparation for upset awareness, prevention, recognition, and recovery.

Academic Material (Knowledge and Risk Management)

Academics establish the foundation for development of situational awareness, insight, knowledge, and skills. As in practical skill development, academic preparation should move from the general to specific while emphasizing the significance of each basic concept. Although academic preparation is crucial and does offer a level of mitigation of the LOC-I threat, long-term retention of knowledge is best achieved when applied and correlated with practical hands-on experience.

The academic portion of UPRT should also address the prevention concepts surrounding aeronautical-decision making (ADM) and risk management (RM), and proportional counter response.

Prevention Through ADM and Risk Management

This element of prevention routinely occurs in a time scale of minutes or hours, revolving around the concept of effective ADM and risk management through analysis, awareness, resource management, and interrupting the error chain through basic airmanship skills and sound judgment. For instance, imagine a situation in which a pilot assesses conditions at an airport prior to descent and recognizes those conditions as being too severe to safely land the airplane. Using situational awareness to avert a potentially threatening flight condition is an example of prevention of an LOC-I situation through effective risk management. Pilots should evaluate the circumstances for each flight (including the equipment and environment), looking specifically for scenarios that may require a higher level of risk management. These include situations that could result in low-altitude maneuvering, steep turns in the pattern, uncoordinated flight, or increased load factors.

Another part of ADM is crew resource management (CRM) or single-pilot resource management (SRM). Both are relevant to the UPRT environment. When available, a coordinated crew response to potential and developing upsets can provide added benefits such as increased situational awareness, mutual support, and an improved margin of safety. Since an untrained crewmember can be the most unpredictable element in an upset scenario, initial UPRT for crew operations should be mastered individually before being integrated into a multi-crew, CRM environment. A crew should be able to accomplish the following:

1. Communicate and confirm the situation clearly and concisely;
2. Transfer control to the most situationally-aware crewmember;
3. Using standardized interactions, work as a team to enhance awareness, manage stress, and mitigate fear.

Prevention Through Proportional Counter-Response

In simple terms, proportional counter-response is the timely manipulation of flight controls and thrust, either as the sole pilot or crew as the situation dictates, to manage an airplane flight attitude or flight envelope excursion that was unintended or not commanded by the pilot.

The time-scale of this element of prevention typically occurs on the order of seconds or fractions of seconds, with the goal being the ability to recognize a developing upset and take proportionally-appropriate avoidance actions to preclude the airplane entering a fully-developed upset. Due to the sudden, surprising nature of this level of developing upset, there exists a high risk for panic and overreaction to ensue and aggravate the situation.

Recovery

Last but not least, the academics portion lays the foundation for development of UPRT skills by instilling the knowledge, procedures, and techniques required to accomplish a safe recovery. The airplane and FSTD-based training elements presented below serve to translate the academic material into structured practice. This can start with classroom visualization of recovery procedures and continue with repetitive skill practiced in an airplane, and then potentially further developed in the simulated environment.

In the event looking outside does not provide enough situational awareness of the airplane attitude, a pilot can use the flight instruments to recognize and recover from an upset. To recover from nose-high and nose-low attitudes, the pilot should follow the procedures recommended in the AFM/POH. In general, upset recovery procedures are summarized in *Figure 5-3*.

Upset Recovery Template
1. Disconnect the wing leveler or autopilot
2. Apply forward column or stick pressure to unload the airplane
3. Aggressively roll the wings to the nearest horizon
4. Adjust power as necessary by monitoring airspeed
5. Return to level flight

Figure 5-3. *Upset recovery template.*

Common Errors

Common errors associated with upset recoveries include the following:

1. Incorrect assessment of what kind of upset the airplane is in
2. Failure to disconnect the wing leveler or autopilot
3. Failure to unload the airplane, if necessary
4. Failure to roll in the correct direction
5. Inappropriate management of the airspeed during the recovery

Roles of FSTDs and Airplanes in UPRT

Training devices range from aviation training devices (e.g., basic and advanced) to FSTDs (e.g., flight training devices (FTD) and full flight simulators (FFS)) and have a broad range of capabilities. While all of these devices have limitations relative to actual flight, only the higher fidelity devices (i.e., Level C and D FFS) are a satisfactory substitution for developing UPRT skills in the actual aircraft. Except for these higher fidelity devices, initial skill development should be accomplished in a suitable airplane, and the accompanying training device should be used to build upon these skills. [Figure 5-4]



Figure 5-4. A Level D full-flight simulator could be used for UPRT.

Airplane-Based UPRT

Ultimately, the more realistic the training scenario, the more indelible the learning experience. Although creating a visual scene of a 110° banked attitude with the nose 30° below the horizon may not be technically difficult in a modern simulator, the learning achieved while viewing that scene from the security of the simulator is not as complete as when viewing the same scene in an airplane. Maximum learning is achieved when the pilot is placed in the controlled, yet adrenaline-enhanced, environment of upsets experienced while in flight. For these reasons, airplane-based UPRT improves a pilot's ability to overcome fear in an airplane upset event.

However, airplane-based UPRT does have limitations. The level of upset training possible may be limited by the maneuvers approved for the particular airplane, as well as by the flight instructor's own UPRT capabilities. For instance, UPRT conducted in the normal category by a typical flight instructor will necessarily be different from UPRT conducted in the aerobatic category by a flight instructor with expertise in aerobatics.

When considering upset training conducted in an aerobatic-capable airplane in particular, the importance of employing instructors with specialized UPRT experience in those airplanes cannot be overemphasized. Just as instrument or tailwheel instruction requires specific skill sets for those operations, UPRT demands that instructors possess the competence to oversee trainee progress, and the ability to intervene as necessary with consistency and professionalism. As in any area of training, the improper delivery of stall, spin, and upset recovery training often results in negative learning, which could have severe consequences not only during the training itself, but in the skills and mindset pilots take with them when they have passengers and place the lives of others at stake.

All-Attitude/All-Envelope Flight Training Methods

Sound UPRT encompasses operation in a wide range of possible flight attitudes and covers the airplane's limit flight envelope. This training is essential to prepare pilots for unexpected upsets. As stated at the outset, the primary focus of a comprehensive UPRT program is the avoidance of, and safe recovery from, upsets. Much like basic instrument skills, which can be applied to flying a vast array of airplanes, the majority of skills and techniques required for upset recovery are not airplane-specific. Just as basic instrument skills learned in lighter and lower performing airplanes are applied to more advanced airplanes, basic upset recovery techniques provide lessons that remain with pilots throughout their flying careers.

FSTD-based UPRT

UPRT can be effective in high fidelity devices (i.e., Level C and D FFS); however, instructors and pilots should be mindful of the technical and physiological boundaries when using a particular FSTD for upset training. This training is a current requirement for pilots seeking a multiengine airplane ATP certificate in accordance with 14 CFR part 61, section 61.156, and the training course must be FAA approved.

Coordinated Flight

Coordinated flight occurs whenever the pilot is proactively correcting for yaw effects associated with power (engine/propeller effects), aileron inputs, how an airplane reacts when turning, and airplane rigging. The airplane is in coordinated flight when the airplane's nose is yawed directly into the relative wind and the ball is centered in the slip/skid indicator (except for certain multiengine airplane operation with an engine failure). [Figure 5-5]

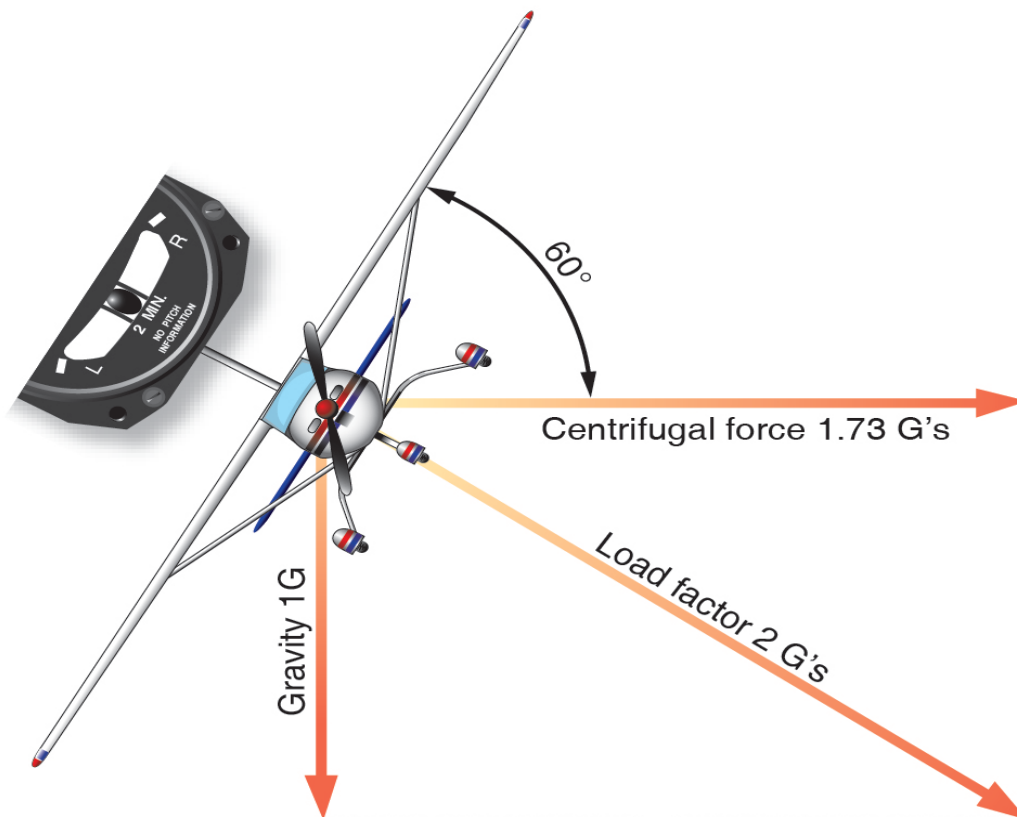


Figure 5-5. Coordinated flight in a turn.

Angle of Attack

The angle of attack (AOA) is the angle at which the chord of the wing meets the relative wind. The chord is a straight line from the leading edge to the trailing edge. At low angles of attack, the airflow over the top of the wing flows smoothly and produces lift with a relatively small amount of drag. As the AOA increases, lift as well as drag increases; however, above a wing's critical AOA, the flow of air separates from the upper surface and backfills, burbles, and eddies, which reduces lift and increases drag. This condition is a stall, which can lead to loss of control if the AOA is not reduced.

It is important for the pilot to understand that a stall is the result of exceeding the critical AOA, not of insufficient airspeed. The term “stalling speed” can be misleading, as this speed is often discussed when assuming 1G flight at a particular weight and configuration. Increased load factor directly affects stall speed (as well as do other factors such as gross weight, center of gravity, and flap setting). Therefore, it is possible to stall the wing at any airspeed, at any flight attitude, and at any power setting. For example, if a pilot maintains airspeed and rolls into a coordinated, level 60° banked turn, the load factor is 2G, and the airplane will stall at a speed that is 41 percent higher than the 1G stall speed. In that 2G level turn, the pilot has to increase AOA to increase the lift required to maintain altitude. At this condition, the pilot is closer to the critical AOA than during level flight and therefore closer to the higher stalling speed. Because “stalling speed” is not a constant number, pilots need to understand the underlying factors that affect it in order to maintain aircraft control in all circumstances.

Slow Flight

Flying at reduced airspeeds is normal in the takeoff/departure and approach/landing phases of flight. While pilots typically perform these operations at low airspeeds and close to the ground, pilots learn to maneuver an airplane in slow flight at a safe altitude. During slow flight, any further increase in angle of attack, increase in load factor, or reduction in power, will result in a stall warning (e.g., aircraft buffet, stall horn, etc.), and pilots should react to and correct for any stall indication. Note that stall training builds upon the knowledge and skill acquired from the slow flight maneuver and encompasses the period of time from the stall warning (e.g., aircraft buffet, stall horn, etc.) to the stall.

The objective of maneuvering in slow flight is to develop the pilot's ability to fly at low speeds and high AOAs. Through practice, the pilot becomes familiar with the feel, sound, and visual cues of flight in this regime, where there is a degraded response to control inputs and where it is more difficult to maintain a selected altitude. It is essential that pilots:

1. understand the aerodynamics associated with slow flight in various aircraft configurations and attitudes,
2. recognize airplane cues in these flight conditions,
3. smoothly manage coordinated flight control inputs while maneuvering without a stall warning, and
4. make prompt appropriate correction should a stall warning occur.

For pilot training and testing purposes, slow flight includes two main elements:

- Slowing to, maneuvering at, and recovering from an airspeed at which the airplane is still capable of maintaining controlled flight without activating the stall warning—5 to 10 knots above the 1G stall speed is a good target.
- Performing slow flight in configurations appropriate to takeoffs, climbs, descents, approaches to landing, and go-arounds.

Slow flight should be introduced with the target airspeed sufficiently above the stall to permit safe maneuvering, but close enough to the stall warning for the pilot to experience the characteristics of flight at a low airspeed. One way to determine the target airspeed is to slow the aircraft to the stall warning when in the desired slow flight configuration, pitch the nose down slightly to eliminate the stall warning, and add power to maintain altitude and note the airspeed.

When practicing slow flight, a pilot learns to divide attention between aircraft control and other demands. How the airplane feels at the slower airspeeds demonstrates that as airspeed decreases, control effectiveness decreases. For instance, reducing airspeed from 30 knots to 20 knots above the stalling speed will result in a certain loss of effectiveness of flight control inputs because of less airflow over the control surfaces. As airspeed is further reduced, the control effectiveness is further reduced and the reduced airflow over the control surfaces results in larger control movements being required to create the same response. Pilots sometimes refer to the feel of this reduced effectiveness as “sloppy” or “mushy” controls.

When flying above the minimum drag speed (L/D_{MAX}), more power is required to fly even faster. When flying at speeds below L/D_{MAX} , more power is required to fly even slower. Since slow flight will be performed well below L/D_{MAX} , the pilot should be aware that large power inputs or a reduction in AOA will be required to prevent the aircraft from decelerating. It is important to note that when flying below L/D_{MAX} or *on the backside of the power curve*, as the AOA increases toward the critical AOA and the airplane's speed continues to decrease, small changes in the pitch control result in disproportionately large changes in induced drag and therefore changes in airspeed. As a result, pitch becomes a more effective control of airspeed when flying below L/D_{MAX} and power is an effective control of the path.

It is also important to note that an airplane flying below L/D_{MAX} , exhibits a characteristic known as “speed instability” and the airspeed will continue to decay without appropriate pilot action. For example, if the airplane is disturbed by turbulence and the airspeed decreases, the airspeed may continue to decrease without the appropriate pilot action of reducing the AOA or adding power. [Figure 5-6]

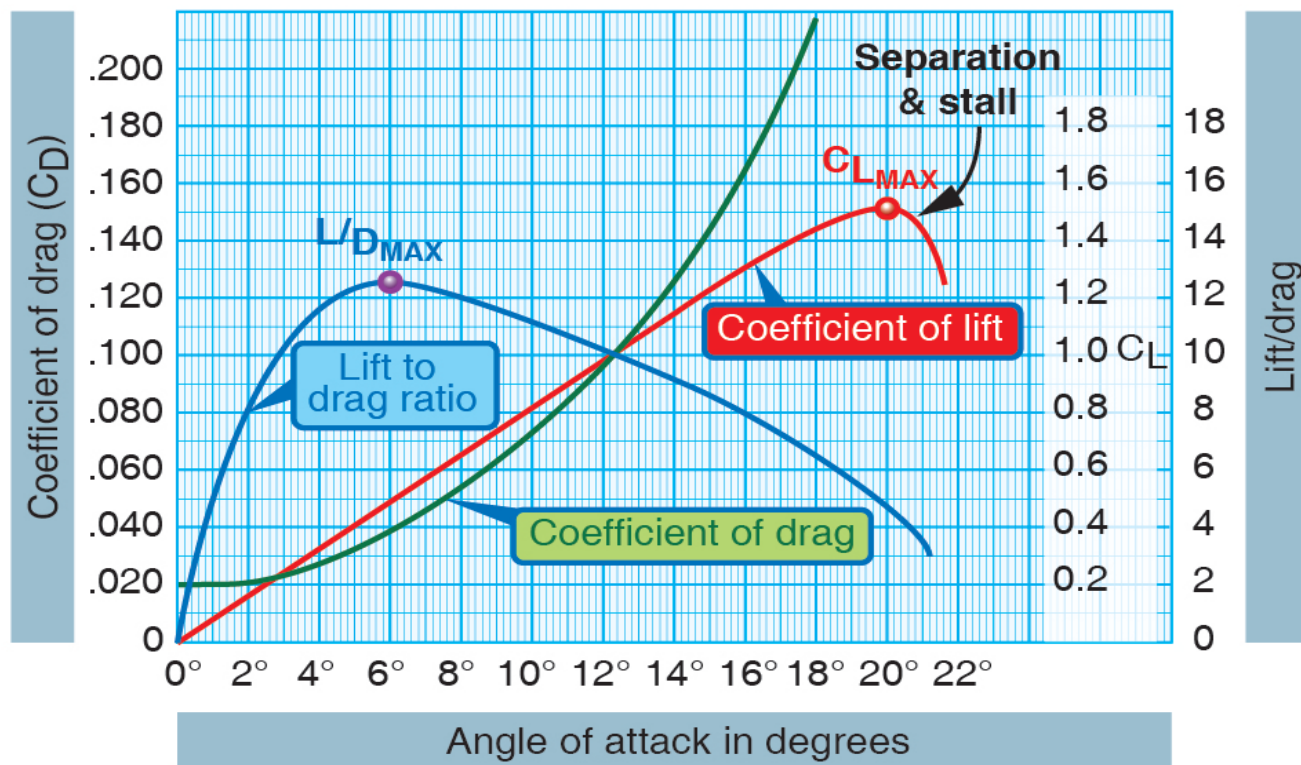


Figure 5-6. Angle-of-attack in degrees.

Performing the Slow Flight Maneuver

Slow flight training includes:

- Slowing the airplane smoothly and promptly from cruising to approach speeds without changes in altitude or heading, while increasing the angle of attack and setting the required power and trim.
- Configuration changes, such as extending the landing gear and adding flaps, while maintaining heading and altitude.
- Turning while maintaining altitude.
- Straight-ahead climbs and climbing medium-banked (approximately 20 degrees) turns, and straight-ahead power-off gliding descents and descending turns, which represent the takeoff and landing phases of flight.

Slow flight in a single-engine airplane should be conducted so the maneuver can be completed no lower than 1,500 feet AGL (3,000 for multiengine airplanes), or higher, if recommended by the manufacturer. In all cases, practicing slow flight should be conducted at an adequate height above the ground for recovery should the airplane inadvertently stall.

To begin the slow flight maneuver, the pilot should clear the area and gradually reduce thrust from cruise power and adjust the pitch to allow the airspeed to decrease while maintaining altitude. As the speed of the airplane decreases, there is a change in the sound of the airflow. As the speed approaches the target slow flight speed, which is an airspeed just above the stall warning in the desired configuration (i.e., approximately 5–10 knots above the stall speed for that flight condition), additional power will be needed to maintain altitude. During these changing flight conditions, the pilot should trim the airplane to compensate for changes in control pressures. If the airplane remains trimmed at the pre-maneuver cruising speed, strong aft (back) control pressure is needed on the elevator, which will make precise control difficult.

Slow flight is typically performed and evaluated in the landing configuration. Therefore, both the landing gear and the flaps should be extended to the landing position, as applicable. It is recommended the prescribed before-landing checks be completed to configure the airplane. The extension of gear and flaps typically occurs once cruise power has been reduced and at appropriate airspeeds to ensure limitations for extending those devices are not exceeded. Practicing this maneuver in other configurations, such as a clean or takeoff configuration, is also good training and may be evaluated on the practical test.

With an AOA just under the AOA which may cause an aerodynamic buffet or stall warning, the flight controls are less effective. [Figure 5-7] The elevator control is less responsive and larger control movements are necessary to retain control of the airplane. In propeller-driven airplanes, torque, slipstream effect, and P-factor may produce a strong left yaw, which requires right rudder input to maintain coordinated flight. The closer the airplane is to the 1G stall, the greater the amount of right rudder pressure required.



Figure 5-7. *Slow flight—low airspeed, high angle of attack, high power, and constant altitude.*

Maneuvering in Slow Flight

When the desired pitch attitude and airspeed have been established in straight-and-level slow flight, the pilot needs to maintain awareness of outside references and continually cross-check the airplane's instruments to maintain control. The pilot should note the feel of the flight controls, especially the airspeed changes caused by small pitch adjustments, and the altitude changes caused by power changes. The pilot should practice turns to determine the airplane's controllability characteristics at this low speed. During the turns, it will be necessary to increase power to maintain altitude. Abrupt or rough control movements during slow flight may result in a stall. For instance, abruptly raising the flaps while in slow flight can cause the plane to stall.

The pilot should also practice climbs and descents by adjusting the power when stabilized in straight-and-level slow flight. The pilot should note the increased yawing tendency at high power settings and counter it with rudder input as needed.

To exit the slow flight maneuver, add power. As airspeed and lift increase, apply forward control pressure to reduce the AOA and maintain altitude. Maintain coordinated flight, level the wings as necessary, and return to the desired flightpath. As airspeed increases, clean up the airplane by retracting flaps and landing gear, if they were extended, and adjust trim as needed. A pilot should anticipate the changes to the AOA as the landing gear and flaps are retracted to avoid a stall.

Common Errors

Common errors in the performance of slow flight are:

1. Failure to adequately clear the area
2. Inadequate back-elevator pressure as power is reduced, resulting in altitude loss
3. Excessive back-elevator pressure as power is reduced, resulting in a climb followed by rapid reduction in airspeed
4. Insufficient right rudder to compensate for left yaw
5. Fixation on the flight instruments
6. Failure to anticipate changes in AOA as flaps are extended or retracted
7. Inadequate power management
8. Inability to adequately divide attention between airplane control and orientation
9. Failure to properly trim the airplane
10. Failure to respond to a stall warning

Stalls

A stall is an aerodynamic condition which occurs when smooth airflow over the airplane's wings is disrupted, resulting in loss of lift. Specifically, a stall occurs when the AOA—the angle between the chord line of the wing and the relative wind—exceeds the wing's critical AOA. It is possible to exceed the critical AOA at any airspeed, at any attitude, and at any power setting. [Figure 5-8]



Figure 5-8. Critical angle of attack and stall.

For these reasons, it is important to understand factors and situations that can lead to a stall, and develop proficiency in stall recognition and recovery. Performing intentional stalls will familiarize the pilot with the conditions that result in a stall, assist in recognition of an impending stall, and develop the proper corrective response if a stall occurs. Stalls are practiced to two different levels:

- **Impending Stall**—an impending stall occurs when the AOA causes a stall warning, but has not yet reached the critical AOA. Indications of an impending stall can include buffeting, stick shaker, or aural warning.
- **Full Stall**—a full stall occurs when the critical AOA is exceeded. Indications of a full stall are typically that an uncommanded nose down pitch cannot be readily arrested, and may be accompanied by an uncommanded rolling motion. For airplanes equipped with stick pushers, their activation is also an indicator of a full stall.

Although it depends on the degree to which a stall has progressed, some loss of altitude is expected during recovery. The longer it takes for the pilot to recognize an impending stall, the more likely it is that a full stall will result. Intentional stalls should therefore be performed at an altitude that provides adequate height above the ground for recovery and return to normal level flight.

Stall Recognition

A pilot should recognize the flight conditions that are conducive to stalls and know how to apply the necessary corrective action. This level of proficiency involves learning to recognize an impending stall by sight, sound, and feel.

Stalls are usually accompanied by a continuous stall warning for airplanes equipped with stall warning devices. These devices may include an aural alert, lights, or a stick shaker all which alert the pilot when approaching the critical AOA. Most vintage airplanes, and many types of light-sport and experimental airplanes, do not have stall warning devices installed. However, certification standards permit manufacturers to provide the required stall warning either through the inherent aerodynamic qualities of the airplane (pre-stall buffeting) or through a stall warning device that gives a clear indication of the impending stall.

Other sensory cues for the pilot include:

- **Feel**—the pilot will feel control pressures change as speed is reduced. With progressively less resistance on the control surfaces, the pilot needs to use larger control movements to get the desired airplane response. The pilot will notice the airplane’s reaction time to control movement increases.
- **Vision**—since the airplane can be stalled in any attitude, vision is not a foolproof indicator of an impending stall. However, maintaining pitch awareness is important.
- **Hearing**—as speed decreases, the pilot should notice a change in sound made by the air flowing along the airplane structure.
- **Kinesthesia**—the physical sensation (sometimes referred to as “seat of the pants” sensations) of changes in direction or speed is an important indicator to the trained and experienced pilot in visual flight. If this sensitivity is properly developed, it can warn the pilot of an impending stall.

Pilots should remember that a level-flight 1G published stalling speed is valid only:

1. In unaccelerated 1G flight
2. In coordinated flight (slip-skid indicator centered)
3. At one weight (typically maximum gross weight)
4. At a particular center of gravity (CG) (typically maximum forward CG)

Angle of Attack Indicators

An AOA indicator gives the pilot better situational awareness pertaining to the aerodynamic health of the airfoil. This can be referred to as stall margin awareness or knowing the existing margin between the current AOA and the critical AOA. While learning to recognize stalls without relying on stall warning devices is important, an AOA indicator provides an additional visual indication of the airplane’s proximity to the critical AOA. The FAA along with the General Aviation Joint Steering Committee (GAJSC) is promoting the use of Angle of Attack (AOA) indicators to reduce the occurrence of loss of control in flight.

Without an AOA indicator, the AOA is “invisible” to pilots. These devices measure several parameters simultaneously and determine the current angle of attack providing a visual image to the pilot of the current AOA along with representation of the proximity to the critical AOA. These devices can give a visual representation of the energy management state of the airplane. The energy state of an airplane is the balance between airspeed, altitude, drag, and thrust and represents how efficiently the airfoil is operating. With this increased situational awareness pertaining to the energy condition of the airplane, the pilot has additional information to help prevent a loss of control scenario.

AOA indicators are increasingly affordable for GA airplanes. There are several different kinds of AOA indicators with varying methods for calculating AOA; therefore, proper installation and training on the use of these devices is important. AOA indicators measure several parameters simultaneously, determine the current AOA, and provide a visual image of the proximity to the critical AOA. [Figure 5-9] Some AOA indicators also provide aural indications, which can provide awareness to a change in AOA that is trending towards the critical AOA prior to installed stall warning systems. It’s important to note that some indicators take flap position into consideration, but not all do.

While AOA indicators provide a simple visual representation of the current AOA and its proximity to the critical AOA, they are not without their limitations. These limitations should be understood by operators of GA airplanes equipped with these devices. Like advanced automation such as autopilots and moving maps, the misunderstanding or misuse of the equipment can have disastrous results. Some items that may limit the effectiveness of an AOA indicator are listed below:

1. Calibration techniques
2. Probes or vanes not being heated
3. The type of indicator itself
4. Flap setting
5. Wing contamination

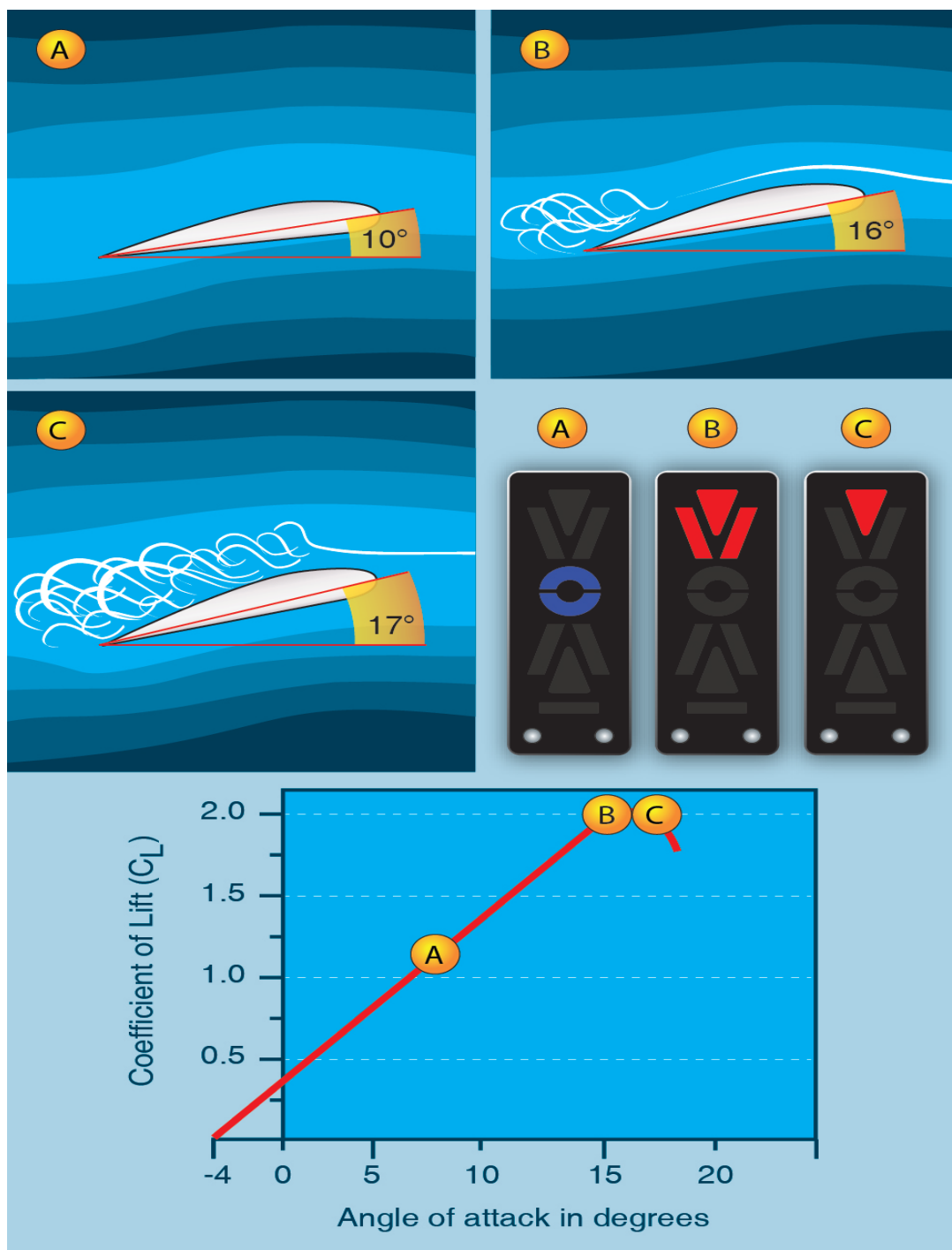


Figure 5-9. A conceptual representation of an AOA indicator. It is important to become familiar with the equipment installed in a specific airplane.

Installation of AOA indicators not required by type certification in GA airplanes has been streamlined by the FAA. The FAA established policy in February 2014 pertaining to non-required AOA systems and how they may be installed as a minor alteration, depending upon their installation requirements and operational utilization, and the procedures to follow for certification of these installations. For updated information on this, please reference the FAA website at www.faa.gov.

If airplane equipment includes an angle of attack indicator, the pilot should know how the particular device determines AOA, what the display indicates, and the appropriate response to any indication. Pilots are encouraged to conduct in-flight training to see the indications throughout various maneuvers, such as slow flight, stalls, takeoffs, and landings, and to practice the appropriate responses to those indications. It is also important to note that some items may limit the effectiveness of an AOA indicator (e.g., calibration techniques, wing contamination, unheated probes/vanes). Pilots flying an airplane equipped with an AOA indicator should refer to the pilot handbook information or contact the manufacturer for specific limitations applicable to that indicator type.

Ground and flight instructors should make every attempt to receive training from an instructor knowledgeable about AOA indicators prior to giving instruction pertaining to or in airplanes equipped with an AOA indicator. Pilot schools should incorporate training on AOA indicators in their syllabi whether their training aircraft are equipped with them or not.

Stall Characteristics

Different airplane designs can result in different stall characteristics. The pilot should know the stall characteristics of the airplane being flown and the manufacturer’s recommended recovery procedures. Factors that can affect the stall characteristics of an airplane include its geometry, CG, wing design, and high-lift devices. Engineering design variations make it impossible to specifically describe the stall characteristics for all airplanes; however, there are enough similarities in small general aviation training-type airplanes to offer broad guidelines.

Most training airplanes are designed so that the wings stall progressively outward from the wing roots (where the wing attaches to the fuselage) to the wingtips. Some wings are manufactured with a certain amount of twist, known as washout, resulting in the outboard portion of the wings having a slightly lower AOA than the wing roots. This design feature causes the wingtips to have a smaller AOA during flight than the wing roots. Thus, the wing roots of an airplane exceed the critical AOA before the wingtips, meaning the wing roots stall first. Therefore, when the airplane is in a stalled condition, the ailerons should still have a degree of control effectiveness until/unless stalled airflow migrates outward along the wings. Although airflow may still be attached at the wingtips, a pilot should exercise caution using the ailerons prior to the reduction of the AOA because it can exacerbate the stalled condition. For example, if the airplane rolls left at the stall (“rolls-off”), and the pilot applies right aileron to try to level the wing, the downward-deflected aileron on the left wing produces a greater AOA (and more induced drag), and a more complete stall at the tip as the critical AOA is exceeded. This can cause the wing to roll even more to the left, which is why it is important to first reduce the AOA before attempting to roll the airplane.

The pilot should also understand how the factors that affect stalls are interrelated. In a power-off stall, for instance, the cues (buffeting, shaking) are less noticeable than in the power-on stall. In the power-off, 1G stall, the predominant cue may be the elevator control position (full up elevator against the stops) and a high descent rate.

Fundamentals of Stall Recovery

Depending on the complexity of the airplane, stall recovery could consist of as many as six steps. Even so, the pilot should remember the most important action to an impending stall or a full stall is to reduce the AOA. There have been numerous situations where pilots did not first reduce AOA, and instead prioritized power and maintaining altitude, which resulted in a loss of control. This section provides a generic stall recovery procedure for light general aviation aircraft adapted from a template developed by major airplane manufacturers and can be adjusted appropriately for the aircraft used. [Figure 5-10] However, a pilot should always follow the aircraft-specific manufacturer’s recommended procedures if published and current.

Stall Recovery Template	
1. Wing leveler or autopilot	1. Disconnect
2. a) Pitch nose-down b) Trim nose-down pitch	2. a) Apply until impending stall indications are eliminated b) As needed
3. Bank	3. Wings Level
4. Thrust/Power	4. As needed
5. Speed brakes/spoilers	5. Retract
6. Return to the desired flight path	

Figure 5-10. Stall recovery template.